Introduction to neural networks

Machine learning recall

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Overview

Machine Learning

A field of study that gives computers the ability to learn without being explicitly programmed [a]

[a] Arthur Samuel, 1959

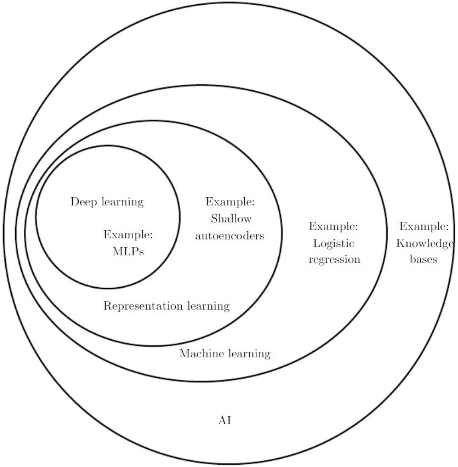
Introduction

Learning

An agent is learning if it improves its performance on future tasks after making observations about the world. [a]

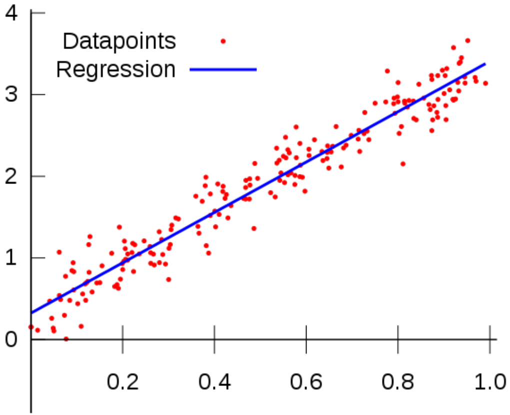
[a] Russell and Norvig

Machine Learning in Artificial Intelligence [1]



[1] Goodfellow, I., Bengio, Y., and Courville, A. (2016). Deep learning. MIT press.

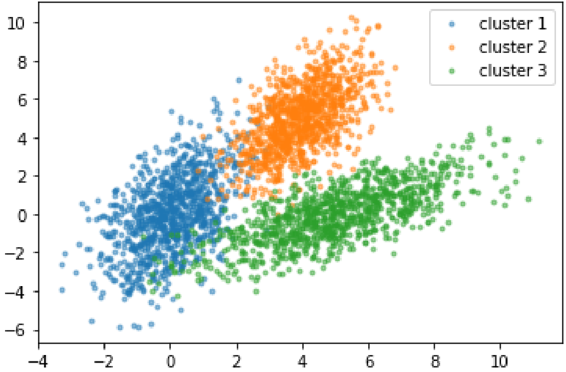
- **Regression.** Finds a function that relates the input to a continuous codomain.



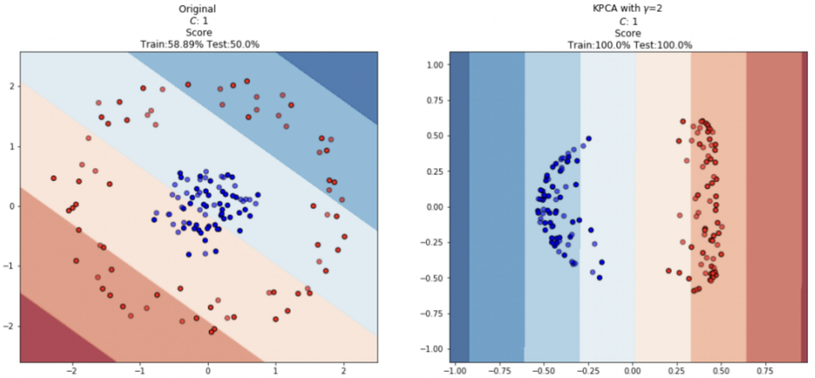
- **Classification.** Assigns a label to a given input. The set of classes is predefined.

or

- **Clustering.** Automatically groups the inputs according to their features. The number of groups may be unknown.



- **Dimensionality reduction.** Describes an entity with less features.

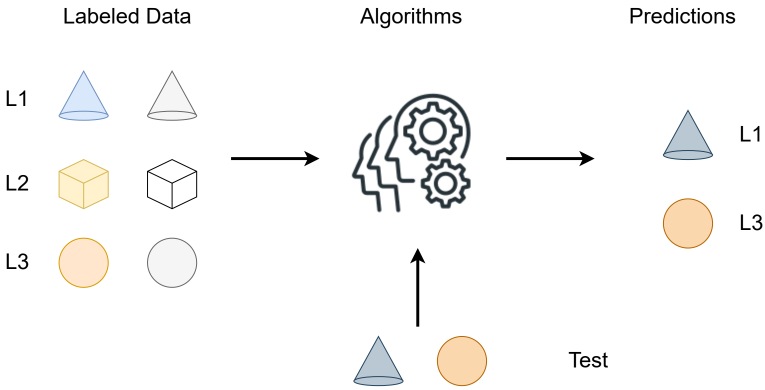


- **Generative modeling.** To create new instances from a distribution.

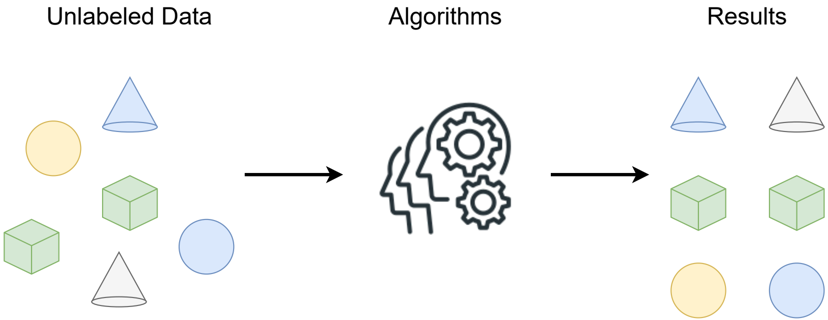
Depending on how machines learn:

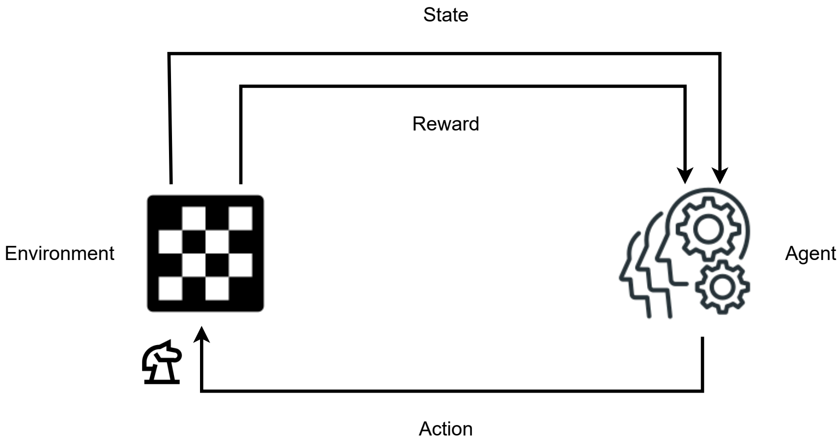
- **Supervised Learning.** The machine learns from labeled data.
- **Un-supervised Learning.** The machine tries to discover the structure of input data without labels.
- **Reinforcement Learning.** The machine interacts with the environment improving its actions.

Supervised Learning



Unsupervised Learning





State
Reward
Environment
Agent
Action

Further readings

- Sections 18.1 and 18.2, *Learning from examples* chapter, Russell and Norvig, Artificial Intelligence: A modern apprach.
- Section 5.1 *Learning Algorithms*. Goodfellow et al., Deep Learning, MIT Press.

References

- https://www.topbots.com/ai-research-generative-adversarial-network-images/
- https://neptune.ai/blog/dimensionality-reduction
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- Photo by charlesdeluvio on Unsplash, Photos: Julie Molliver Shane Rounce